JOSHUA ZENG

Principal Software Engineer | DevOps and Automation | Active TS Clearance

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Career Experience

Principal Software Engineer, Northrop Grumman *Oct 2019 – Present*

- Accelerate product readiness and reduce integration risk for 5 teams with CI/CD/CT pipelines and security scans in Jenkins, TeamCity and Bamboo, using IaC (Ansible, Terraform, Packer) and Jenkins SaaS cloud agents
- Served as lead developer of the Nexus migration, authoring the Python scripts and libraries that moved all images and artifacts and tracked migration status across Artifactory and Nexus (including npm)
- Lead operations and customer liaison for critical internal tooling and software suites, keeping them automated and serviceable for multiple engineering teams (Atlassian administration, artifact repositories, custom CI/CD)
- Created capacity-monitoring and health-check tools (C#, Selenium, Jaspersoft) that flag Atlassian instances nearing limits (e.g., custom fields) and deviations from naming, workflow and other administration standards
- Maintain internal test automation tooling in Python and Flask, with C#/.NET installers and Docker deployment; deliver WPF/WinForms apps for skip-level leaders and managers
- Headed automated Jira reporting that compiles story points, summaries, labels and other issue data into a Word document of each project's scope and completed work for managers
- Developed automated contract code-count reports with Grafana and AWS-loaded Postgres entity-relationship models
- Spearheaded the modernization of a legacy automated pilot-training test suite to current SIL testing standards, integrating it with a closed-loop aircraft simulator

Senior Software Engineer, Driiven.ai *Jan – Mar 2026*

- Implemented full-stack features for an ad customer platform displaying margin and financial report metrics

Software Engineer, BAE Systems *Aug 2018 – Oct 2019*

- Co-led the architectural migration of an iSCSI server from C# to C++ using multithreading and networking
- Developed a Windows packet comparison utility (C#/.NET) and a C++ port security app with PnP drivers
- Designed the front-end UI/UX for hardware and digital circuit design of an armament tester, per armament specifications

Software Engineer, Qualcomm via Manpower *Mar – Aug 2018*

- Engineered a C#/.NET desktop suite for camera Auto Focus and Auto Exposure Control tuning, pairing a backend that organizes tuning data with a front end of reports and visualizations for hardware calibration teams
- Developed a lens-movement convergence simulation that maps tuning metrics and charts targets against raw data, with an integrated XML editor for tuning camera parameters and faster tuning cycles

Camera Test Engineer, Google via Adecco *Summers 2016 and 2017*

- Created a Python color metric analysis app (OpenCV, wxPython) that auto-detects Macbeth charts for camera color accuracy
- Wrote a color metric tuning parser in C with an interface to directly adjust many tuning parameters the codebase lacked
- Conducted 3A KPI image-quality tests, LED, video (EIS) and power tests, and used simulation and tuning software to identify camera bugs, including hard-artifact scenes

Education

M.S., Computer Science (GPA 4.0/4.0), Georgia Institute of Technology *Aug 2020 – Dec 2025*

B.S., Electrical and Computer Engineering (ML/AI specialization), UC San Diego *Sept 2014 – June 2018*

Skills and Certifications

Languages & frameworks: Python, C#, C++, C, Java, JavaScript, HTML/CSS; Flask, Django, .NET, WPF, WinForms, Selenium, Android Studio, MATLAB

DevOps, tooling and reporting: Git (GitHub, GitLab), Jenkins, TeamCity, Bamboo, Docker, IaC (Ansible, Terraform, Packer), Artifactory, Nexus, Atlassian admin, Grafana, PostgreSQL, AWS, SonarQube, Jaspersoft, Tableau

Hardware: Arduino, Raspberry Pi, SOLIDWORKS, Fusion, oscilloscopes and signal generators

Certifications: CompTIA Security+; National Instruments CLAD